1 Theory

Lensing formalism
Lectures on gravitational lensing (Narayan and Bartelmann)
galaxiesbook

1.1 Basic theory

Gravitational lensing is one of the most famous predictions from Einstein's theory of general relativity. A massive object will curve spacetime and light will follow this curvature, so this means the path of light will be bent. This means that depending on the alignment of the source, lens and observer, lensing can produce images of background objects which look warped into rings, stretched shapes and multiple images.Generally, people categorise lensing events into:

- Strong lensing: large masses with favourable geometries. We see two or more separate images of the source
- Weak lensing: large masses with less favourable geometry. we see speared or sheared images
- Microlensing: involves small masses where we can't resolve the images from one another. We instead see a brightening of the source

[diagram here?]

For a light ray passing near mass M, general relativity predicts a deflection angle

$$\alpha = \frac{4GM}{c^2 b}$$

where b is the impact parameter.

The lens equation relates the true position of the background object to the apparent position of the lensed image to the mass distribution of the lens.

$$\beta = \theta - \frac{D_{LS}}{D_S}\alpha(\theta)$$

In the case where the source, lens and observer lie along a straight line, the lens equation predicts a circular image called and einstein ring. Otherwise, we get 2 images.

General relativity and Liouville's theorem say that gravitational lensing can't change the surface brightness of an object (i.e. its brightness per unit area on the sky.) When an object is lensed, the image appers magnified. The image covers a larger angular area and each bit of that area has the same surface brightness as before so the total flux increases. We can define the magnificationmu as

$$\mu = \frac{\text{observed flux}}{\text{intrinsic flux}}$$

It might feel like this violates enegry conservation, or that the lens creates light but this is not the case. The lesns bends light rays towards us that would have missed us, so we recieve more of the light that was already being emitted. In a microlensing event, our images are too close together for us to reasonably be able to separate them with a telescope, so we see a change in brightness over time. We see a total flux increase called a lightcurve. The magnification for a point lens is

$$\mu(u) = \frac{u^2+2}{u\sqrt{u^2+4}}$$

where u is the angular separation between lens and source in terms of the Einstein radius.

1.2 Images and Caustics

The lens equation is

$$\beta = \theta - \alpha(\theta)$$

It essentially connects two planes: θ - the apparent angular positoin of an image on the sky and the source plane (where the object actually is)
β - the true position of a source with no lensing
α - deflection caused by the gravitational lens
This connection (mapping) is non linear, so creates many things such as arcs, rings and multiple images. The equation has two solutions for θ for a point mass which is why a point mass will produce two images, even in microlensing where we can't resolve them. The **Jacobian** of lens mapping is

$$A = \frac{\partial \beta}{\partial \theta}$$

which we can derive from the lens equation if we really wanted to. We can find the magnification from

$$\mu = \frac{1}{\det A}$$

so the magnification becomes infinite when the determinant is 0. Curves in the image plane where det(A) =0 are called critical curves. If we map these critical curves through the lens equation into the source plane the resulting curves are called caustics. Near the caustics, the amplification is descirbed by Airy (fold) or Pearcey (cusp) integrals

2 Interstellar Communication

Electromagnetic signals spread out as they travel, with the recieved power dropping as $1/r^2$. At interstellar distances this means that we're looking at signals becoming trillions of times weaker, a massive hurdle for interstellar communication. Not only would they be hard to detect, but they would be difficult to pick out from background noise. To overcome this by modifying our transmitters alone would require either massive city-spanning transmitters or huge amounts of energy. If we could use the sun to amplify transmissions, we could not only increase the power of our signal but also focus it into a narrow beam.

2.1 The Sun

The minimum allowed impact parameter is set by the physical radius r_0 of the lensing body. The minimal focal distance z_{min} is

$$z_{min} = \frac{r_0^2}{2g}$$

where

$$g = \frac{2GM}{c^2}$$

is the gravitational radius. If we susbtitute in values for the sun we get $z_{min} =$ 550AU. This is the minimum start of the focal line. Perfect alignment gives us an einstein ring. For a coherent signal, the sun focuses the wavefrin into a diffraction-limited spot. Ray optics predicts that the intensity near the optical axis (line of perfect alignment) will grow as

$$I \propto \frac{1}{\sqrt{x^2/(2gz)}}$$

The divergence is cut off at the Fresnel scale so the smalled spot a lens can form is

$$x_F = \sqrt{\frac{\lambda z}{2\pi}}$$

The gravitational equivalent of the Airy disc. Plug this in to our intensity equation gives

$$I_{max} = 2\pi k g$$

where $k = 2\pi/\lambda$. So for radio wavelengtrhs we get signal boosts in the area of 10 million.

2.2 The Corona

The sun's corona also bends radio waves and will bend them in the opposite direction to our gravitational bending. The plasma bending is dominant at radio frequencies. It also introduces random phase shifts into the wavefront which would destroy the coherent amplification needed for the huge gain we mentioned previously. To avoid this we must use short wavelengths i.e millimeter or sub-millimeter.